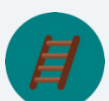


3.5 Applying Exponent Laws

Lesson Introduction

 Benchmark	MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational-number bases, with procedural fluency.		 Learning Target	I can apply the laws of exponents to generate equivalent numerical expressions.
 Benchmark Coherence	Building On MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational bases.		Working Toward MA.912.NSO.1.1 Extend previous understanding of the laws of exponents to include rational exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.	
 Essential Questions	- How do you rewrite a rational base that has a negative exponent as an equivalent exponential expression with a positive exponent? - How do you apply multiple exponent laws to generate equivalent numerical expressions?			
 Lesson Narrative	This lesson extends students' work with the laws of exponents to include rational-number bases. Students explore how the exponent impacts the value of the numerical expressions with fractional bases. Students then collaboratively practice using multiple exponent laws to generate equivalent numerical expressions involving rational bases and integer exponents. Through the activities, students gain experience with using different laws and/or a different sequence in applying exponent laws to generate an expression that is equivalent. Students individually demonstrate their conceptual understanding and procedural fluency after several activities that involve reflective practice.			
 Component Summary	3.5.1 Warm-Up (~5 min.)	Select the expression with larger value and justify (<i>SE p. 124</i>)	3.5.4 Collaboration (10 min.)	Identify if two expressions are equivalent and use the laws of exponents to justify (<i>SE p. 127</i>)
	3.5.2 Guided Instruction (10-15 min.)	Explore patterns in equivalent expressions that have a rational base and exponent (<i>SE p. 125</i>)	3.5.5 Your Turn! (5-10 min.)	Rewrite numerical expressions to have one base and a positive exponent using the laws of exponents (<i>SE p. 128</i>)
	3.5.3 Collaboration (10 min.)	Analyze and complete student work for using exponent laws to write equivalent expressions (<i>SE p. 126</i>)	3.5.6 Wrap-Up (~5 min.)	Demonstrate mastery of learning target (<i>SE p. 129</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> Negative exponent law		(Optional) Highlighters	
		Additional Preparation Use observational data from prior lessons to plan for strategic grouping to use during the lesson.		

 Teacher Tips	Common Misconceptions - When working with negative fractions, students may incorrectly take the opposite instead of the reciprocal. For example, rewriting $\left(\frac{-2}{3}\right)^{-5}$ as $\left(\frac{2}{3}\right)^5$. - Students may not understand why rational bases with positive exponents are equivalent to their reciprocals with negative exponents. For example: $\left(\frac{2}{3}\right)^1 = \left(\frac{3}{2}\right)^{-1}$. - Students may believe that the laws of exponents must be applied in a specific order. 3.5.3 Collaboration reinforces for students that laws can be applied in a different order and result in the same value.	Things to Consider - Prompt students to write the expanded form to check their application of the laws of exponents. - Discuss how the patterns recognized in 3.5.2 Guided Instruction connect to multiplying and dividing fractions. - Review how to convert decimals to their equivalent fractions. - Expressions can be simplified by applying the laws of exponents in different orders. During debrief discussions, ask students who applied the laws in different sequences to share their work.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Number Talks 3.5.1 Warm-Up offers an opportunity to incorporate a Number Talk by asking a variety of students to share and justify their answers. Furthermore, throughout 3.5.2 Guided Instruction, there are more opportunities to use this routine to discuss patterns students notice and how the laws of exponents can be applied to numerical expressions with rational bases.	MA.K12.MTR.5.1: Patterns and Structures In 3.5.2 Guided Instruction, students discover how to apply the laws of exponents to write equivalent expressions for numerical expressions with rational bases. Students identify patterns they notice as the exponent increases and decreases with numerical expressions with rational bases. They consolidate this understanding as they discover how to apply the negative law of exponents.
 Differentiate Instruction	3.5.3 Collaboration provides students with two models of how to describe a plan for applying multiple exponent laws to write equivalent expressions. Provide students with sentence stems as they answer the questions in 3.5.3 Collaboration. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

3.5.1 Warm-Up: Math Argument

Component Overview: Students select which of the given expressions has a greater value. Students justify their selection.



3.5.1 Warm-Up: Math Argument

- Without performing any calculations, circle the expression you think has the greater value.

$$60^{63}$$

$$63^{60}$$

- Justify why you selected that expression.

Answers will vary. The exponent of the first expression is larger (63).



Prompt students to consider a simpler problem and then apply the same logic to this problem. For example: Does 6^3 or 3^6 result in a greater value?



Does this hold for all rational numbers a and b , when $a < b$, then $a^b > b^a$? For example, which expression has the greater value? $\left(\frac{1}{2}\right)^6$ or $6^{\left(\frac{1}{2}\right)}$?

3.5.2 Guided Instruction: Rational Number Bases

Component Overview: Students are guided through how to apply the negative exponent law to write equivalent expressions for numerical expressions with rational bases.



3.5.2 Guided Instruction: Rational Number Bases

1. Consider the table shown.

Exponential Expression	Expanded Form	Equivalent Expression
$\left(\frac{2}{5}\right)^3$	$\frac{2}{5} \cdot \frac{2}{5} \cdot \frac{2}{5}$	$\frac{2^3}{5^3}$
$\left(\frac{2}{5}\right)^2$	$\frac{2}{5} \cdot \frac{2}{5}$	$\frac{2^2}{5^2}$
$\left(\frac{2}{5}\right)^1$	$\frac{2}{5}$	$\frac{2^1}{5^1}$
$\left(\frac{2}{5}\right)^0$	1	$\frac{2^0}{5^0}$



Consider describing the expressions verbally to provide an entry point for auditory learners. Also, provide an entry point for visual learners by modeling fraction division and multiplication using number lines or area model.

Complete the statement to describe the pattern in the table.

As the value of the exponent ☐ increases ☒ decreases by 1 in the column on the left, the equivalent expression in the column on the right can be found by ☐ multiplying ☒ dividing the previous value by the ☐ base. ☒ exponent.



Consider discussing with students how the patterns in the tables in this lesson compare to the patterns in the tables in the previous lesson. When you evaluate a rational base to a positive exponent, is the value greater or less than the rational base? How does this connect to the rules for multiplying and dividing fractions?

3.5.2 Guided Instruction: Rational Number Bases (continued)

Continuing the pattern, the next row in the table would begin with the expression $\left(\frac{2}{5}\right)^{-1}$.

To continue the pattern, find the value of $1 \div \frac{2}{5}$. Since dividing by a fraction is equivalent to multiplying by the fraction's reciprocal, the result is as follows.

$$1 \div \frac{2}{5} = 1 \cdot \frac{5}{2} = \frac{5}{2}$$

Since $\left(\frac{2}{5}\right)^{-1} = \frac{5}{2}$ and the identity exponent law states $\frac{5}{2} = \left(\frac{5}{2}\right)^1$, it can be concluded that $\left(\frac{2}{5}\right)^{-1} = \left(\frac{5}{2}\right)^1$.

2. Continue the pattern based on this information.

Exponent	Expanded Form	Equivalent Expression
$\left(\frac{2}{5}\right)^0$	1	$\frac{2^0}{5^0}$
$\left(\frac{2}{5}\right)^{-1}$	$\frac{5}{2}$	$\left(\frac{5}{2}\right)^1$
$\left(\frac{2}{5}\right)^{-2}$	$\frac{5}{2} \cdot \frac{5}{2}$	$\left(\frac{5}{2}\right)^2$
$\left(\frac{2}{5}\right)^{-3}$	$\frac{5}{2} \cdot \frac{5}{2} \cdot \frac{5}{2}$	$\left(\frac{5}{2}\right)^3$



What is the reciprocal of a fraction $\frac{a}{b}$, where a, b are integers? What is the difference between taking the reciprocal and taking the opposite?

Explain if the expressions $\left(\frac{2}{3}\right)^1$, $\left(\frac{3}{2}\right)^{-1}$ are equivalent.

3. Complete the statement.

A rational base with a negative exponent, $\left(\frac{a}{b}\right)^{-m}$, can be

rewritten using the ☐ reciprocal ☐ opposite of the base with the

☐ reciprocal ☐ opposite exponent.

3.5.2 Guided Instruction: Rational Number Bases (continued)

4. Re-write each expression as an equivalent exponential expression using positive exponents.

a. $\left(\frac{2}{3}\right)^{-4} = \left(\frac{3}{2}\right)^4$

b. $\left(\frac{3}{-5}\right)^{-9} = \left(\frac{-5}{3}\right)^9$

c. $\left(-\frac{8}{9}\right)^{-2} = \left(-\frac{9}{8}\right)^2$

d. $\left(\frac{21}{4}\right)^{-5} = \left(\frac{4}{21}\right)^5$



Consider having students highlight, circle, and/or draw arrows to keep track of the changes they are making as they apply the negative exponent law.

Consider having students take turns verbally explaining rewriting the expressions to their partner. The partner should perform the steps explained by their partner to rewrite the expression. Then for the next expression, the partners switch roles.

3.5.3 Collaboration: Applying More Than One Exponent Law

Component Overview: Students analyze two different students' approaches to writing an equivalent numerical expression using more than one exponent law. Students describe and discuss which method they prefer.



3.5.3 Collaboration: Applying More Than One Exponent Law

1. Fernanda and Junior were asked to write an equivalent expression of $\left(\frac{2.5^5}{2.5^3}\right)^{-1}$ by applying multiple exponent laws. Each student first described their plan.
 - a. Complete each student's work based on the description of their plan.

Fernanda	Junior
I will first apply the quotient of powers exponent law. Then, I will apply the power of a power exponent law. Finally, I will re-write the decimal as a fraction and apply the negative exponent law.	I will first apply the power of a quotient exponent law. Then, I will apply the quotient of powers exponent law. Finally, I will re-write the decimal as a fraction and apply the negative exponent law.
$\begin{aligned} &\left(\frac{2.5^5}{2.5^3}\right)^{-1} \\ &= (2.5^2)^{-1} \\ &= (2.5)^{-2} \\ &= \left(\frac{5}{2}\right)^{-2} \\ &= \left(\frac{2}{5}\right)^2 \end{aligned}$	$\begin{aligned} &\left(\frac{2.5^5}{2.5^3}\right)^{-1} \\ &= \frac{(2.5)^{-5}}{(2.5)^{-3}} \\ &= (2.5)^{-2} \\ &= \left(\frac{5}{2}\right)^{-2} \\ &= \left(\frac{2}{5}\right)^2 \end{aligned}$

- b. Explain which method you prefer.
Answers will vary. I prefer Fernando's method because he simplifies the fraction in the second step.
 - c. Ask your partner which method they prefer. Summarize your partner's explanation.
Answers will vary. I prefer Junior's method because he distributed the -1 into the fraction, and then applied the quotient rule.



How do you convert a decimal to an equivalent fraction?
Which exponent law was used at each step?



Consider providing sentence stems for this prompt:

- "The student's method (or plan) I prefer is..."
_____.
- "I prefer this method (or plan) because..."
_____.
- "One thing I didn't like about the other method was..."
_____.
- "My partner preferred _____'s method. My partner preferred this method because..."
_____.

3.5.4 Collaboration: Equivalent Expressions

Component Overview: Students identify which expressions are equivalent and use the laws of exponents to justify their answers.



3.5.4 Collaboration: Equivalent Expressions

1. Consider the expression $\frac{(\frac{1}{6})^3}{6^{-5}} \cdot 6^4$.

For each expression, work with your partner to determine if it is equivalent to $\frac{(\frac{1}{6})^3}{6^{-5}} \cdot 6^4$. Then, justify your choice by explaining using laws of exponents or showing your work.

Expression	Is it Equivalent to $\frac{(\frac{1}{6})^3}{6^{-5}} \cdot 6^4$?	Justify or Explain
a. $\frac{6^{-3}}{6^{-5}} \cdot 6^4$	☐ Yes ☐ No	$(\frac{1}{6})^3 = 6^{-3}$
b. $6^{-8} \cdot 6^4$	☐ Yes ☐ No	Applying the quotient of powers law to $\frac{6^{-3}}{6^{-5}}$ results in 6^2 , which is not equal to 6^{-8}
c. $(\frac{1}{6})^{-2} \cdot 6^4$	☐ Yes ☐ No	Applying the quotient of powers law to $\frac{6^{-3}}{6^{-5}}$ results in 6^2 , which is equivalent to $(\frac{1}{6})^{-2}$.
d. $(\frac{1}{6})^{-8} \cdot 6^4$	☐ Yes ☐ No	Applying the quotient of powers law to $\frac{6^{-3}}{6^{-5}}$ results in 6^2 , which $\neq (\frac{1}{6})^{-8}$.
e. $\frac{(\frac{1}{6})^{-3}}{(\frac{1}{6})^5} \cdot (\frac{1}{6})^{-4}$	☐ Yes ☐ No	$6^{-3} \neq (\frac{1}{6})^{-3}$
f. 6^6	☐ Yes ☐ No	$\frac{6^{-3}}{6^{-5}} \cdot 6^4 = 6^2 \cdot 6^4 = 6^6$



Consider using Sage and Scribe as a strategy. Have a student share their answer as if the expressions are equivalent. Then, have them explain or justify verbally while their partner takes notes on the exponent laws they used to determine if they are equivalent. Then have partners switch roles.

To add complexity, when partners switch roles, the new sage can be asked to explain if the expressions are equivalent using a different order or different series of laws applied.

3.5.5 Your Turn!

Component Overview: Students independently practice applying the laws of exponents to match and generate equivalent expressions.



3.5.5 Your Turn!

1. Match each expression in Column A to an equivalent expression in Column B by drawing arrows.

Column A	Column B
$(-\frac{7}{8})^{-2}$	$(\frac{7}{8})^2$
$(-\frac{8}{7})^{-2}$	$(-\frac{8}{7})^2$
$(\frac{7}{8})^{-2}$	$(\frac{8}{7})^2$
$(\frac{8}{7})^{-2}$	$(-\frac{7}{8})^2$

2. Three expressions are shown.
Circle the two expressions that are equivalent.

$$\left(\frac{5}{8}\right)^{-2} \cdot \left(\frac{5}{8}\right)^4$$

$$\frac{\left(\frac{5}{8}\right)^{-2}}{\left(\frac{5}{8}\right)^{-4}}$$

$$\left(\left(\frac{5}{8}\right)^{-2}\right)^4$$

3. Apply the laws of exponents to write an equivalent numerical expression with one base and one positive exponent. Show your steps in the column on the left and identify the law of exponents used in each step in the column on the right.

$$\left[\left(\frac{2}{3}\right)^5 \cdot \left(\frac{2}{3}\right)^{-2}\right]^3$$

Steps	Laws of Exponents
$\left(\frac{2}{3}\right)^{15} \cdot \left(\frac{2}{3}\right)^{-6}$	Power of a product
$\left(\frac{2}{3}\right)^9$	Product of powers



On question 1, some students may match different expressions than in the key, including cases of one expression from column A to more than one expression from column B. For example, a student might notice that $(\frac{7}{8})^2$ and $(-\frac{7}{8})^2$ are equivalent expressions as well because they have the same value. In cases where students mark different expressions, consider asking students to explain their reasoning to determine whether there is a misconception about exponent laws or if they are recognizing the equivalence of the values.



Why is the third expression in question 2 not equivalent to the first and second expressions? How could you revise the expression to make it equivalent?



For students who finish early, consider asking them to complete problem 3 using a different sequence of steps.

3.5.6 Wrap-Up: Ticket Out the Door

Component Overview: Students demonstrate their mastery of the lesson learning target. Use data from this formative assessment to determine which students need remediation.



3.5.6 Wrap-Up: Ticket Out the Door

1. Rewrite each expression as an equivalent exponential expression with positive exponents.

a. $\left(\frac{-13}{2}\right)^{-3} = \left(\frac{2}{-13}\right)^3$

b. $\left(\frac{1}{5}\right)^{-8} = \left(\frac{5}{1}\right)^8 = 5^8$

2. Generate an equivalent expression using the laws of exponents. Express your answer with positive exponents.

$$\left(\frac{-5}{4}\right)^{-7} \cdot \left(\frac{-5}{4}\right)^{-3}$$

$$\left(\frac{4}{-5}\right)^7 \cdot \left(\frac{4}{-5}\right)^3 = \left(\frac{4}{-5}\right)^{10}$$



The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.



Proficiency on this 3.5.6 Wrap-Up includes (1) fluency in applying the negative exponent law and (2) fluency in using more than one exponent law to write an equivalent expression for a numeric expression with a rational base and negative exponent.